

Comparing Physiological and Emotional Effects of Happy and Sad Virtual Environments Experienced in Video and Virtual Reality

Yuankun Zhu^{1*}Arindam Dey^{1†}¹University of Queensland

ABSTRACT

Virtual Reality (VR) could give users a more immersive experience than other non-immersive mediums. In this study, we explored differences in emotional and physiological effects between videos and VR using two different sets of contents to evoke happy and sad emotions. In this within-subjects controlled experiment we collected real-time heart rate, positive and negative affect schedule (PANAS), and Self-Assessment Manikin (SAM) to measure physiological and emotional effects. Our results showed that VR triggers stronger emotions and higher heart rate than videos.

1 INTRODUCTION

Virtual Reality (VR) is an immersive medium that can provide high-quality realism and interaction akin to the real world and beyond. It has been an effective medium in evoking emotions [3]. It has also been argued that emotion can affect the feeling of presence in a virtual environment (VE) [4]. Dey et al. have showed that, by utilizing real-time heart rate feedback, VR can communicate emotional states between the users in a collaborative application [2] and it can also manipulate emotion in a single-user setup [1].

These studies have used a visualization of a physiological signal to passively communicate emotion. In our work, we are interested in detecting emotions in real-time using physiological signals such as heart rate and galvanic skin response while experiencing immersive VEs and then share that emotion directly to the users or using it to adapt the VEs to suits user's emotional needs. Earlier work exists to measure real-time emotions during exposure to video stimuli [5] using physiological signals, however, such research is not available for VR. As a first step towards understanding the differences in emotional and physiological effects between videos and VR, we performed this preliminary exploration. This results can also be useful in measuring the differences of emotional arousal in commonly used VR social platform such as Virbela and VRChat, and their 2D counterparts that are akin to videos.

To this effort, we have designed a *within-subjects* user study with 20 participants to measure the differences of same environments experienced through video on a screen and immersive VR. We investigated this differences on two sets of environments evoking happy and sad emotions. Our results pointed to significant differences.

2 USER STUDY

The Virtual Environment, Procedure, & Equipment: We designed two different environments – Happy and Sad. The happy environment exposed participants to a bright vivid forest with blue lakes, flying birds, and a mountain in the backdrop (Figure 1a). This environment was supplemented by appropriate sounds effects including a happy ambient music. On the contrary, the sad environment exposed participants to a burning forest with many dead and



(a) happy



(b) sad

Figure 1: Experimental Environments - (a) happy and (b) sad.

crying animals (Figure 1b). There was appropriate sounds effects to supplement the scene. In both of the environments, participants followed a fixed path and there was no opportunity to interact with the environment. Both environments lasted for four minutes. The VR environment was experienced through a HTC Vive Pro device and the video was experienced through a 15" monitor with a wireless headphone. In both of the experiences participants were sitting on a comfortable revolving chair and were not holding the controller. The screen was placed at an arms distance from the participants but they were allowed to re-position themselves to see the screen comfortably. They were allowed to rotate their head. The heartrate data was collected using an Empatica E4 band.

We designed a *within-subjects* study where all participants experienced the same environments in both Video and VR mediums. We counterbalanced the order of mediums and environments using a Latin-square. After experiencing each environment participants had to fill out Positive and negative affect Schedule (PANAS) and Self-Assessment Manikin (SAM) questionnaires that followed by a 3 minute break before getting exposed to the next environment. The entire study took on average one hour per participant.

Participants: We recruited 20 participants (mean age = 24.78) including 11 males and 9 females. All of them were university students with normal vision and no mental illness. Nine of the participants never experienced VR before.

Hypothesis: We hypothesised that VR will result in higher values in terms of heart rate, PANAS, and SAM scales.

*e-mail: yuankun.zhu@uq.net.au

†e-mail: a.dey@uq.edu.au

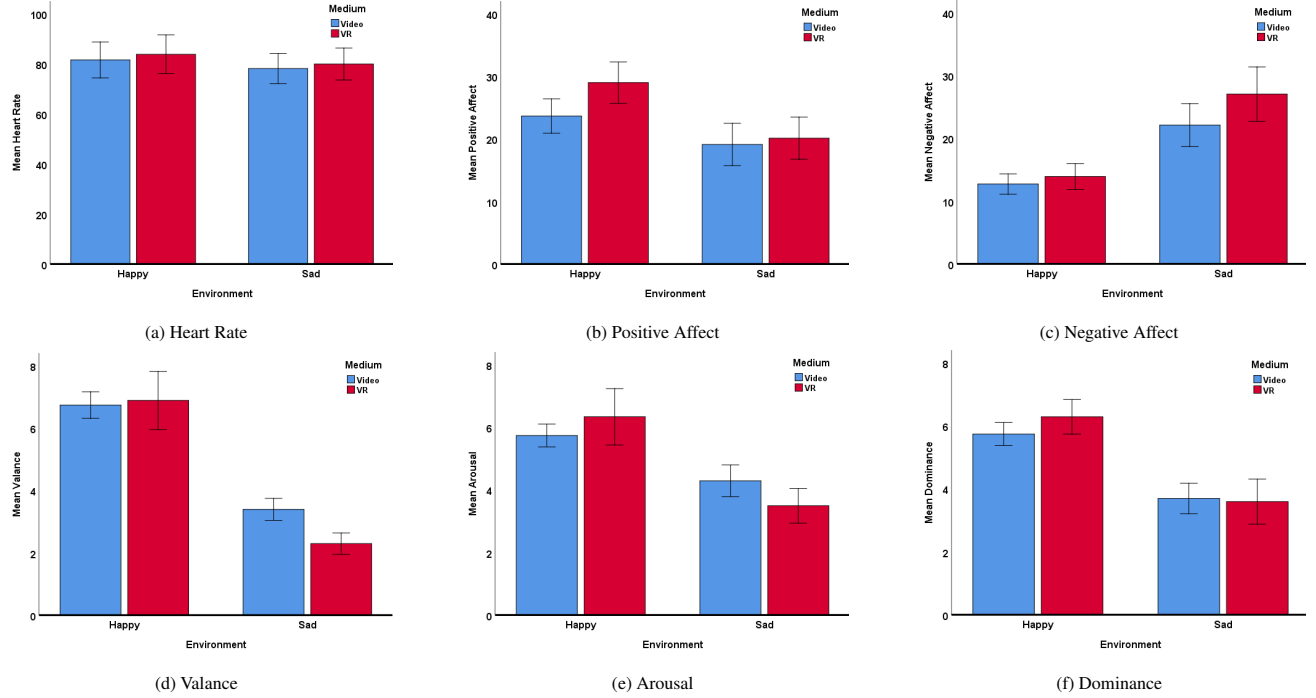


Figure 2: Results: (a) Heart Rate, (b) Positive Affect, (c) Negative Affect, (d) Valance, (e) Arousal, and (f) Dominance. Blue bars = Video, Red Bars = VR. Whiskers represent $\pm 95\%$ confidence intervals.

3 RESULTS & DISCUSSION

To analyse the data we used 2×2 (Medium $\times$ Environment) repeated measure ANOVAs for each of the three dependent variables.

We did not notice any significant main effect of either of the independent variable on **heart rate** (Figure 2a) although VR caused slightly higher heart rate ($p=.08$). It is not surprising as these VEs were non-interactive and experienced while sitting on a chair.

We noticed a significant main effect of environment— $F(1,19) = 21.4, p < .001, \eta_p^2=.53$ —and medium— $F(1,19) = 25.6, p < .001, \eta_p^2=.57$ —on **positive affect** (PA) where the happy VE and VR resulted in higher PA (Figure 2b). We have also noticed a significant interaction effect— $F(1,19) = 11.86, p = .003, \eta_p^2=.38$ —where the difference in PA was between the two mediums was higher for the happy VE than the sad VE.

Similar main effects were noticed on **negative affect** (NA) with VR resulting in higher NA than video— $F(1,19) = 59.7, p < .001, \eta_p^2=.78$ —and sad VE resulting higher NA than happy VE— $F(1,19) = 24.6, p < .001, \eta_p^2=.57$ (Figure 2c). A significant interaction effect— $F(1,19) = 15, p = .001, \eta_p^2=.44$ —showed that VR amplified NA more than video in the sad VE.

We noticed a significant main effect of environment— $F(1,19) = 142.2, p < .001, \eta_p^2=.88$ —and medium— $F(1,19) = 11.1, p = .004, \eta_p^2=.37$ —on **valance** (Figure 2d) where happy VE and video caused higher valance than sad VE and VR respectively. There was a significant interaction effect— $F(1,19) = 6.9, p = .02, \eta_p^2=.26$ —showing that the difference in valance between the mediums was more prominent in sad VE.

We noticed a main effects of environment— $F(1,19) = 48.7, p < .001, \eta_p^2=.72$ —on **arousal** (Figure 2e) with happy VE causing higher arousal than sad VE. A significant interaction effect— $F(1,19) = 6.9, p = .02, \eta_p^2=.27$ —was noticed where VR caused higher arousal for happy VE but lower arousal for sad VE than video.

We noticed a significant main effect of environments on **domi-**

nance— $F(1,19) = 47.1, p < .001, \eta_p^2=.71$ where happy VE caused higher dominance than sad VE (Figure 2f).

4 CONCLUSION

We compared between VR and video mediums used to experience a happy and a sad VE in terms of heart rate and emotions. We noticed that VR intensifies both positive and negative affects than videos. It causes less valance and arousal than videos in sad VEs. Overall, our results show clear yet complex differences between VR and videos in terms of emotional activation. It highlights the need for more dedicated research to measure real-times emotions in VR as the earlier results from video-based research cannot be directly used in VR.

REFERENCES

- [1] A. Dey, H. Chen, M. Billinghamurst, and R. W. Lindeman. Effects of manipulating physiological feedback in immersive virtual environments. In *Proceedings of the 2018 Annual Symposium on Computer-Human Interaction in Play*, pp. 101–111, 2018.
- [2] A. Dey, H. Chen, C. Zhuang, M. Billinghamurst, and R. W. Lindeman. Effects of sharing real-time multi-sensory heart rate feedback in different immersive collaborative virtual environments. In *2018 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, pp. 165–173. IEEE, 2018.
- [3] A. Felnhöfer, O. D. Kothgassner, M. Schmidt, A.-K. Heinze, L. Beutl, H. Hlavacs, and I. Kryspin-Exner. Is virtual reality emotionally arousing? investigating five emotion inducing virtual park scenarios. *International journal of human-computer studies*, 82:48–56, 2015.
- [4] C. Jicol, C. H. Wan, B. Doling, C. H. Illingworth, J. Yoon, C. Headey, C. Lutteroth, M. J. Proulx, K. Petrini, and E. O'Neill. *Effects of Emotion and Agency on Presence in Virtual Reality*. Association for Computing Machinery, New York, NY, USA, 2021.
- [5] J. Shukla, M. Barreda-Angeles, J. Oliver, G. Nandi, and D. Puig. Feature extraction and selection for emotion recognition from electrodermal activity. *IEEE Transactions on Affective Computing*, 2019.